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UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION
SCHOOL OF EDUCATION AND LEADERSHIP
COLLEGES OF EDUCATION
END OF SEMESTER ONE EXAMINATIONS FOR LEVEL 200, 2024/2025
B.ED. PROGRAMME

COURSE CODE: TEJS 221
COURSE TITLE: BIOPHYSICAL ENVIRONMENTAL SYSTEMS

Instruction: Answer **all** questions in Section A and any **three** questions in Section B.
Time: 2 hours

SECTION A

[25 Marks, One (1) Mark for Each Question]
Answer all the questions in this section in the answer booklet.

1. The phenomenon of rising global temperatures due to increasing greenhouse gases in the atmosphere is known as:
   A. Global cooling
   B. Global warming
   C. Ozone depletion
   D. Acid rain
2. The breaking down of rocks into smaller pieces without chemical changes is called?
   A. Chemical weathering
   B. Mechanical weathering
   C. Deposition
   D. Transportation
3. The dropping of detached material/block from a cliff-face downslope is referred to as?
   A. Rockfall
   B. Landslide
   C. Earth flow
   D. Debris flow
4. The point inside the Earth where an earthquake originates is called:
   A. Epicenter
   B. Focus
   C. Fault line
   D. Crater

5. The four major realms of the Earth include:
   A. Lithosphere, Hydrosphere, Atmosphere, Biosphere
   B. Mountains, Rivers, Valleys, Oceans
   C. Troposphere, Stratosphere, Mesosphere, Thermosphere
   D. Core, Mantle, Crust, Outer Core
6. The realm of the Earth that contains all living organisms is called:
   A. Lithosphere
   B. Biosphere
   C. Atmosphere
   D. Hydrosphere
7. The layer of gases surrounding the Earth is known as the:
   A. Lithosphere
   B. Hydrosphere
   C. Atmosphere
   D. Stratosphere
8. Seismic waves that travel through the earth's interior are called
   A. Surface waves
   B. Rayleigh waves
   C. Love waves
   D. Body waves
9. Which of the following is the outermost layer of the internal structure of the earth?
   A. Biosphere
   B. Lithosphere
   C. Core
   D. Crust
10. Savanna vegetation type in Ghana is primarily characterised by:
   A. Dense tall trees with a closed canopy
   B. Evergreen trees with high biodiversity
   C. A mix of grasses and scattered trees often adapted to seasonal drought
   D. Coniferous trees and shrubs with needle-like leaves
11. The process by which water changes from liquid to gas and enters the atmosphere from a body of water is known as?
   A. Condensation
   B. Precipitation
   C. Evaporation
   D. Infiltration
12. Ghana experiences two main rainy seasons; Which of these geographical factors is most responsible for this pattern?
   A. Presence of the Volta Lake
   B. Harmattan winds
   C. Seasonal movement of Inter-Tropical Convergence Zone (ITCZ)
   D. Proximity of coastline to the Atlantic Ocean

13. Which of the following is the primary driving force behind folding in layers of rock?
    A. Sudden intense heat
    B. Plate divergence
    C. Volcanic eruptions
    D. Plate convergence
14. Which type of volcano is characterised by broad, gently sloping sides and is built up by successive flows of basaltic lava?
    A. Composite volcano
    B. Cinder cone
    C. Shield volcano
    D. Caldera
15. Which of these forces is responsible for the formation of reverse faulting.
    A. Tensional
    B. Compressional
    C. Earthquake
    D. Fault
16. The largest ocean on Earth is the:
    A. Atlantic Ocean
    B. Indian Ocean
    C. Pacific Ocean
    D. Arctic Ocean
17. All the following are examples of extrusive volcanic features **except…**
    A. Caldera lake
    B. Dyke
    C. Lava plateau
    D. Geyser
18. Vibrations of the earth's crust due to the movement of the earth's plate is referred to as?
    A. Faulting
    B. Folding
    C. Volcanicity
    D. Earthquake
19. Dark clouds extending to heights of about 15,000 metres which bring a lot of rain to tropical lands are known as ……..
    A. Cirrocumulus
    B. Cirrus
    C. Cumulonimbus
    D. Cumulus
20. Which of the following is **NOT** an example of an exogenic process?
    A. Weathering
    B. Erosion
    C. Deposition
    D. Mantle convection

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21. What type of rainfall occurs when warm air rises over a mountain?
    A. Orographic rainfall
    B. Convectional rainfall
    C. Cyclonic rainfall
    D. Acid rain
22. Sequential development of distinctive soils along a slope is known as?
    A. Soil catena
    B. Soil profile
    C. Soil texture
    D. Soil consistency
23. The force that cause winds to move from one locality to another is called?
    A. Coriolis force
    B. Friction
    C. Gravity
    D. Pressure gradient
24. Which of the following are planetary winds?
    A. NE trade winds
    B. Sea breeze
    C. Land breeze
    D. Wind gusts
25. Which of the following factors will not impede the growth of vegetation?
    A. Soil pollution
    B. Overgrazing
    C. Waterlogging
    D. Sparse vegetal cover

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SECTION B

[75 Marks. Each Question is twenty-five (25) Marks]

Answer any three questions in this section.

Answer three questions from this Section. All questions carry equal marks

1. a. What is earthquake? (4 marks)
   b. Describe four causes of earthquakes? (12 marks)
   c. Discuss three effects of earthquake on man. (9 marks)

2. Explain the earth's realms and their relationships. (25 marks)

3. a. Describe the following folds:
   i. Simple fold [3 marks]
   ii. Asymmetrical fold [3 marks]
   iii. Over fold [3 marks]
   b. Using appropriate diagrams, state ***two*** characteristics and briefly describe formation of the following:
   i. Block Mountains [5 marks]
   ii. Rift Valleys [5 marks]
   iii. Fold Mountains [6 marks]

4. a. Define the term soil erosion [2 marks]
   b. Explain how ***three*** of the following promote soil erosion:
   i. Shifting cultivation
   ii. Over grazing
   iii. Surface mining
   iv. Deforestation
   v. Construction of roads
   [5 marks × 3 = 15 marks]
   c. Explain ***four*** effects of mass wasting [8 marks]

5. Discuss the factors that influence the growth of vegetation (25 marks)